MAT 127A Discussion 1 Notes

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§I Warm-Up

> Suppose that $x \in \mathbb{Z}$. If $x^2 - 6x + 5$ is even, show that x is odd.

Proof. Suppose for the sake of contradiction that x is even, so $x = 2k$ for some $k \in \mathbb{Z}$. Then, $x^2 - 6x + 5$ would be odd because

$$\begin{aligned} x^2 &= (2k)^2 = 4k^2 = 2(2k^2) && \text{is even,} \\ -6x &= -6(2k) = -12k = 2(-6k) && \text{is even,} \\ 5 &= 2(2) + 1 && \text{is odd,} \end{aligned}$$

and indeed

$$x^2 - 6x + 5 = 2(2k^2 - 6k + 2) + 1 \text{ is odd.}$$

This contradicts the assumption, so x must be odd.

§II Proof by Contradiction

Here is a comparison between direct proof and proof by contradiction.

<table>
<tr><th></th><th>If A, show B</th><th>Show that an $x \in X$ such that … exists</th></tr>
<tr><td>Direct proof</td><td>Since A,
then….
Thus, B.</td><td>Let $x = $ ….
We need to show that x satisfies ….
…</td></tr>
<tr><td>Proof by contradiction</td><td>Suppose for the sake of contradiction that B is false,
then….
This is a contradiction, so B must be true.</td><td>Suppose for the sake of contradiction that all $x \in X$ satisfies the negation of …,
then….
This is a contradiction, so there exists an $x \in X$ that satisfies ….</td></tr>
<tr><td colspan="3">

Does there exist a coin showing a tail?
If we suppose all coins are showing heads and arrive at a contradiction,
then there must exist a coin showing a tail.</td></tr>
<tr><td>Proof by contrapositive</td><td>It suffices to prove the contrapositive that B is false implies A is false.
Suppose B is false,
then….
Thus, A is false.</td><td>N/A</td></tr>
</table>

Remark 1: The proof that we used in the warm-up was branded as a proof by contradiction, but it can also be restated as a proof by contrapositive.

Remark 2: Other important proof techniques that we need to know include **proof by induction** and **proof by cases**.

§III Injective and Surjective Functions

Recall that a function $f : A \to B$ is injective if

$$\text{for all } x_1, x_2 \in A \text{ such that } f(x_1) = f(x_2), \text{ we have } x_1 = x_2.$$

Remark: To show that f is **not injective** amounts to showing the existence of some $x_1, x_2 \in A$ such that $f(x_1) = f(x_2)$, but we have $x_1 \neq x_2$. This characterization is useful to internalize because going from "f is not injective" to this characterization requires a proof by contradiction/contrapositive. Internalizing it reduces this mental load.

A function $f : A \to B$ is surjective if

$$\text{for all } y \in B, \text{ there exists an } x \in A \text{ such that } f(x) = y.$$

> (Lecture 3) Let $X = \{x_1, x_2, ..., x_n\}$. If $f : X \to X$ is surjective, show that it is injective.

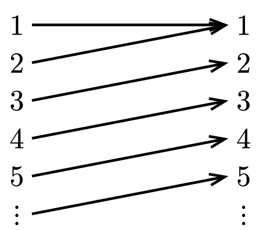
Proof. It suffices to prove the contrapositive that if f is not injective, then it is not surjective. Since f is not injective, the subset $\{f(x_1), f(x_2), ..., f(x_n)\}$ has strictly fewer than n elements. Therefore, there exists some element in X that cannot be mapped to, so f is not surjective.

A function is bijective if it is both injective and surjective.

> (Exercise 1.2.8 modified) Give an example of each. If we have time, try to write down a mathematical proof for each of them.
> (a) A function $f : \mathbb{N} \to \mathbb{N}$ that is surjective but not injective.
> (b) A function $f : \mathbb{N} \to \mathbb{N}$ that is injective but not surjective.
> (c) A function $f : \mathbb{Z} \to \mathbb{N}$ that is bijective.

Sample solution.

(a) $f(1) = 1$ and $f(x) = x - 1$ for $x \geq 2$.



- It is surjective because for all $y \in \mathbb{N}$, we can let $x = y + 1$, then we have

$$f(x) = f(y + 1) = (y + 1) - 1 = y.$$

- It is not injective because $f(1) = 1$ and $f(2) = 1$ but $1 \neq 2$.

(b) $f(x) = x + 1$.

$$\begin{tikzcd}[row sep=tiny]
1 \arrow[dr] & 1 \\
2 \arrow[dr] & 2 \\
3 \arrow[dr] & 3 \\
4 \arrow[dr] & 4 \\
5 \arrow[dr] & 5 \\
\vdots & \vdots
\end{tikzcd}$$

- It is injective because $f(x_1) = f(x_2)$ means $x_1 + 1 = x_2 + 1$, which implies $x_1 = x_2$.
- It is not surjective because there does not exist an $x \in \mathbb{N}$ such that $f(x) = 1$ (*i.e.* $x + 1 = 1$).

(c) $f(x) = 2x$ for $x \geq 1$ and $f(x) = -2x + 1$ for $x \leq 0$.

$$\begin{array}{l}
\left.\begin{array}{rcl}
\vdots & \longrightarrow & \vdots \\
-4 & \longrightarrow & 9 \\
-3 & \longrightarrow & 7 \\
-2 & \longrightarrow & 5 \\
-1 & \longrightarrow & 3 \\
0 & \longrightarrow & 1
\end{array}\right\} \text{Odd natural numbers} \\
\left.\begin{array}{rcl}
1 & \longrightarrow & 2 \\
2 & \longrightarrow & 4 \\
3 & \longrightarrow & 6 \\
4 & \longrightarrow & 8 \\
\vdots & \longrightarrow & \vdots
\end{array}\right\} \text{Even natural numbers}
\end{array}$$

- It is injective because if $f(x_1) = f(x_2)$, then we have the following cases:
  - If both $x_1, x_2 > 0$, then $2x_1 = 2x_2$ implies $x_1 = x_2$.
  - If both $x_1, x_2 \leq 0$, then $-2x_1 - 1 = -2x_2 + 1$ implies $x_1 = x_2$.
  - The case $x_1 > 0$ and $x_2 \leq 0$ cannot happen because $f(x_1) = 2x_1$ is even while $f(x_2) = -2x_2 + 1$ is odd.
  - Similarly, the case $x_1 < 0$ and $x_2 \geq 0$ cannot happen.

  In all possible cases, we have $x_1 = x_2$, so f is injective.
- It is surjective because for all $y \in \mathbb{N}$, we have the following cases:
  - If $y \in \mathbb{N}$ is even, then let $x = \frac{y}{2}$. Since $x > 0$ (because $y > 0$), we have

$$f(x) = 2x = 2\left(\frac{y}{2}\right) = y.$$

  - If $y \in \mathbb{N}$ is odd, then let $x = -\frac{y-1}{2}$. Since $x \leq 0$ (because $y - 1 \geq 0$), we have

$$f(x) = -2x + 1 = -2\left(-\frac{y-1}{2}\right) + 1 = y.$$

  Therefore, we can always find an $x \in \mathbb{Z}$ such that $f(x) = y$, so f is surjective.